

ROKHAN AHMADZAI

Henrietta, NY 14467 | (585) 404-8285 | rokhanahmadzai5@gmail.com

LinkedIn: [linkedin.com/in/rokhan-ahmadzai](https://www.linkedin.com/in/rokhan-ahmadzai) | Portfolio: bit.ly/tigerrit

TECHNICAL SKILLS

- **Software & Tools:** SolidWorks (CAD), MATLAB, AutoCAD, Quartus Prime, C++, Python, Pasco Capstone.
- **Engineering:** FEA (Stress/Strain), GD&T, Design for Manufacture and Assembly (DFMA), Structural Analysis, Autonomous Robotics Test & Build, Digital Systems Logic.
- **Materials & Fabrication:** Aluminum Alloys, Composites, 3D Printing (Slicing/Prototyping), Hand Tools, Soldering.
- **Languages:** English (Professional), Dari & Pashto (Native), Urdu & Hindi (Professional Speaking).

EDUCATION

Rochester Institute of Technology (RIT) | Rochester, NY | Expected May 2029 *Masters of Science (M.S.) in Mechanical Engineering* | **Status:** Accepted for Fall 2026 enrollment with 3rd-year engineering standing.

Monroe Community College (MCC)

| Rochester, NY | Expected May 2026 *Associate in Science (A.S.) in Engineering Science & Physics*

- **Coursework:** Calculus III, Dynamics, Mechanics of Materials, MATLAB, Engineering Physics II.
- **Honors:** 1st Prize Trophy (Mechanics Statics), Matt Hachee Award for Engineering Promise, 1st Place (Class) / Top 3 (College-Wide) Engineering Design & Prototyping.

RESEARCH & TECHNICAL PROJECTS

Research Lead, Iron Eagle (Autonomous Robot) | Spring 2026

- Reduced autonomous aluminum chassis weight by 15% using SolidWorks optimization and DFMA, confirming structural integrity with FEA stress analysis.
- Programmed C++ navigation parameters enabling real-world autonomous pathfinding and environment interaction.

Lead Analyst, Mechanics Statics Structural Design & Physics | Fall 2025

- Modeled a Ferris wheel equilibrium system in PASCO Capstone, aligning theoretical and experimental stress data.
- Built a MATLAB Physics Toolbox to streamline kinematics/dynamics problem-solving and validate theoretical models.

Lead Developer, Autonomous & Digital Systems (Engineering 161/157) | Fall 2024

- Engineered a Python-based autonomous navigation system with sensor logic for reliable robotic object manipulation
- Designed microwave control logic in Quartus Prime and deployed zero-defect operations on FPGA hardware.

PROFESSIONAL EXPERIENCE

Technical Support & Operations | Startek / Northland Controls | 2022 – 2024

- Resolved complex software network failures for a 15,000+ customer base under strict SLAs using root cause analysis
- Monitored global security infrastructure in a fast-paced GSOC, executing real-time incident response protocols.

Peer Mentor | TRIO Student Support Services | Jan 2025 – Aug 2025

Improved retention of first-generation students by developing collaborative study strategies and tracking progress metrics

Work-Study Assistant | MCC Libraries | Jan 2024 – Present

- Spearheaded new catalog systems for critical research, streamlining resource access for 8,000+ students.

LEADERSHIP & COMMUNICATION

Head of Household | Family Relocation & Logistics | 2021 – Present

- Led an emergency family relocation from Afghanistan to the US, managing crisis logistics and finances

Invited Speaker | Academic & Diplomatic Discourse | 2018 – Present

- Delivered addresses on cross-cultural discourse and technical knowledge sharing at RIT, MCC, and Model UN, leveraging fluency in 5 languages.